

Systemic Risk Prognosis via Machine Learning in Networks: A theoretical perspective

Thiago Christiano Silva,*

Benjamin Miranda Tabak**

Abstract

Much has been done in the financial stability literature to describe and understand drivers contributing to systemic risk buildup. The economy interconnectedness is a natural feature that has gained increased importance after the global financial crisis. Motivated by the high integration and globalization of markets worldwide, we have seen that local events can develop into widespread systemic events through networks. Financial network models are essentially descriptive and only look at the past. This paper takes an innovative approach and attempts to construct a model to predict systemic risk using a mixture of econometric models and data-driven techniques borrowed from machine learning. Our forward-looking method can estimate the entire financial network using link prediction techniques and heuristics extracted and trained from past data. Cross-validation methodologies can train the predictor and give robustness to the estimates. During the training phase, we propose a novel way to compare the estimates to the ground-truth networks using graph kernels. In the same vein as the inflation dynamics model, this paper provides novel insights on conducting policies to ensure financial stability.

Keywords: systemic risk, machine learning, network, forecasting, link prediction, graph kernel.

JEL Classification: C63; G01; G20; G21; G28; O16; O40.

*Universidade Catolica de Brasilia

**Getulio Vargas Foundation, School of Public Policy and Government (FGV/EPPG), e-mail: benjaminm.tabak@gmail.com.

1 Introduction

Since the advent of the financial crisis of 2008-2009, regulators and the financial stability literature have sought to build systemic risk measures that incorporate the interconnectivity among financial institutions. These interconnections form a complex network in which economic agents maintain a series of relations among themselves as loans, derivative operations, obligations, among others. Most articles calculate systemic risk using the network of interdependencies among financial institutions. In this way, it is possible to evaluate the amplification effect due to contagion generated by the network and how adverse shocks spread to a set of institutions, leading to greater financial fragility.

We seek to contribute to this debate by developing a methodology to estimate future systemic risk. While most articles use the current or past interconnection networks as given—i.e., they are backward-looking—our methodology is forward-looking. We propose a framework that can estimate the *entire* network in the future using a mixture of robust link prediction techniques borrowed from the machine learning literature and dynamic panel-data econometrics models. The availability of this future network paves the way for regulators to conduct forward-looking systemic risk analysis, which has not been explored in the related literature.

It is important to emphasize that our methodology does not compete with methods that attempt to estimate systemic risk and is quite general. We are proposing an abstract way of estimating a future complex network. This estimated network can be used as input in models of estimation of systemic risk and measurement of amplification of adverse shocks and contagion. In this way, it is possible to adapt any stress measures to the estimated network.¹

To our knowledge, this is the first article that tackles the problem of estimating the future network of interdependencies among financial institutions. While the model takes into account particularities of interbank networks—such as the existence of a core-periphery topology and sparseness and disassortative mixing properties—it is still very general and can be used in other applications, including the estimation of future trade firm-firm networks (purchases and sales) or even payment systems networks. Nevertheless, our method constitutes an important advance in the network-based financial stability literature in what concerns how to look at systemic risk and is extremely useful for the development of monitoring tools.

Perhaps one of the main problems in dealing with the estimation of complex networks is how to calibrate the model. In our case, the model outputs an entire network. A non-trivial question during model selection is: how can we compare the estimated network with the ground-truth network? We tackle this problem by using graph kernels, which incorporate not only local aspects of the graph (direct neighborhood) but also more distant structural properties (indirect neighborhood or neighbors of the neighbor). Our approach departs from the traditional approach of comparing graphs using network descriptors, such as the average degree, strength, or centrality of network vertices, which are very weak because we can obtain totally different networks and still respect these constraints. This is because complex networks have nontrivial connectivity patterns that would be lost in the case of aggregation. In

¹The methodology is sufficiently general and can be applied in other contexts in which it is necessary to estimate a complex network that involves different types of transactions between economic agents.

contrast, graph kernels incorporate these individual connectivity patterns and therefore are more robust².

Since the seminal article of Eisenberg and Noe (2001), which developed an algorithm that allows the clearing of a financial system and provides information on the systemic risk that financial institutions face, several articles have developed methods to calculate the systemic risk of financial systems. At present, there are already several indicators of systemic risk that allow assessing the risks inherent to the financial system from different perspectives, and that can be extremely useful for the development of macro-prudential policies and systemic risk mitigation.

Relevant examples are Rogers and Veraart (2013), which extend Eisenberg and Noe (2001)'s algorithm by introducing default costs if the loans need to be called in by a bank during bankruptcy, and Csóka and Jean-Jacques Herings (2018), which relaxes the assumption of a centralized clearing payment matrix and shows that, in decentralized clearing payment setups, the final value of equity is the same as in a centralized clearing payment procedure, under mild conditions.

Recent papers have proposed the use of different algorithms to evaluate systemic risk. Bayesian methods (Gandy and Veraart (2017)), threat indices to measure spillover effects that may stem from interdependencies between financially linked institutions (Demange (2018)), a combination of methods from risk management and network of interbank loans (Elsinger et al. (2006)), systemic risk measures that can be decomposed in the risk due to individual agents (Chen et al. (2013)), systemic risk with a feedback approach (Silva et al. (2017a), Souza et al. (2016)), DebtRank (Battiston et al. (2012a), Battiston and Martinez-Jaramillo (2018), Battiston et al. (2012b), Tasca et al. (2017))³.

These methods can measure potential risks, systemic risk, likelihood of contagion, and amplification of adverse shocks. However, they do so in the current or past network. Our estimated network allows a forward-looking assessment of potential risks while also taking advantage of this rich ongoing literature on systemic risk indicators. That is, any of these measures can be used in the estimated network. In this way, our contribution is especially relevant for thinking about the development of public policies that seek to mitigate possible risks in a proactive manner. Moreover, it explicitly considers the interconnectivity among economic agents, a feature that is absent in traditional models of econometric predictions.

Contributions to the literature are numerous. First, we present a methodology that estimates a complex network of interactions between multiple agents that somehow transact with each other. Second, we have constructed a way to compare graphs using graph kernels, which is quite complex and still poorly understood in the literature. Third, we innovate by proposing the use of a mix of link prediction, machine learning, and complex networks to estimate a network.

²For a discussion on the use of machine learning in statistical physics and several interesting applications, see Decelle (2023), Wang and Zong (2023), Li et al. (2022), and Tabak et al. (2021).

³See also Acemoglu et al. (2015), Elliott et al. (2014), Nier et al. (2007), Capponi and Chen (2015), Gai and Kapadia (2010), and Erol and Ordoñez (2017)).

2 Methodology

In this paper, we propose a novel framework to perform systemic risk prognoses using network-based machine learning and econometric methods. Section 2.1 standardizes notation used in the paper. Section 2.2 presents an overview of the proposed framework. The subsequent sections detail each step of the framework.

2.1 Notation

We denote $\mathcal{N}$ as the set of participant banks and $N = |\mathcal{N}|$, the number of banks. Moreover, $\mathbf{X}(t) \in \mathcal{N} \times \mathcal{N}$ denotes the matrix of bilateral gross exposures (total assets inside the network)⁴ at time $t \in \mathbb{N}_*$. The (i, j) -th entry, $\mathbf{X}_{ij}(t) \in [0, \infty)$, $(i, j) \in \mathcal{N} \times \mathcal{N}$, represents the exposure of bank i to bank j at time t . $\mathbf{X}(t)$ is a weighted directed graph. The adjacency matrix of $\mathbf{X}(t)$, denoted here as $\mathbf{A}(t)$, can be obtained with the following non-linear transformation:

$$\mathbf{A}_{ij}(t) = \begin{cases} 1, & \text{if } \mathbf{X}_{ij}(t) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The interbank total assets of bank i at instant t are $A_i(t) = \sum_{j=1}^N \mathbf{X}_{ij}(t)$ and its interbank total liabilities at time t are $L_i(t) = \sum_{j=1}^N \mathbf{X}_{ji}(t)$. The interbank total assets and liabilities vectors are denoted, respectively, as $\mathbf{A}(t), \mathbf{L}(t) \in N \times 1$. Whenever the context is clear, we drop the time indexing t so as not to clutter the notation. In this case, we assume we are dealing with an arbitrary $t \in \mathbb{N}_*$.

2.2 A quick overview of the framework

The systemic risk prognosis framework accepts as input the set of interbank network snapshots at time $t \in \{1, \dots, T\}$, where T is the instant of the current or last known snapshot of the interbank network. We use such a set of interbank networks as the *training set* in a supervised learning setting to train our model. Given the training set, the goal is to predict the interbank network at $t = T + 1$, which is unknown and denoted as the *test set*.

Figure 1 elucidates how the overall prognosis process is conducted. Succinctly, there are two main processes that generate guidelines and constraints that serve as inputs for the network estimation model: (a) the construction of the link probability map and (b) the prediction of aggregate network-based descriptors for each bank in the network. The main idea of each step is as follows:

- *Prediction of the link probability map:* given the trajectory of the interbank networks for $t \in \{1, \dots, T\}$, we use a link prediction algorithm to estimate the existence or absence of links in the network at a posterior time $T + 1$. This estimated probability matrix informs the existence likelihood of links between any pairs of banks (vertices) in the network. To prevent the prediction

⁴In several countries, netting between participants is not legally enforceable. In this way, we use the gross rather than liquid exposures.

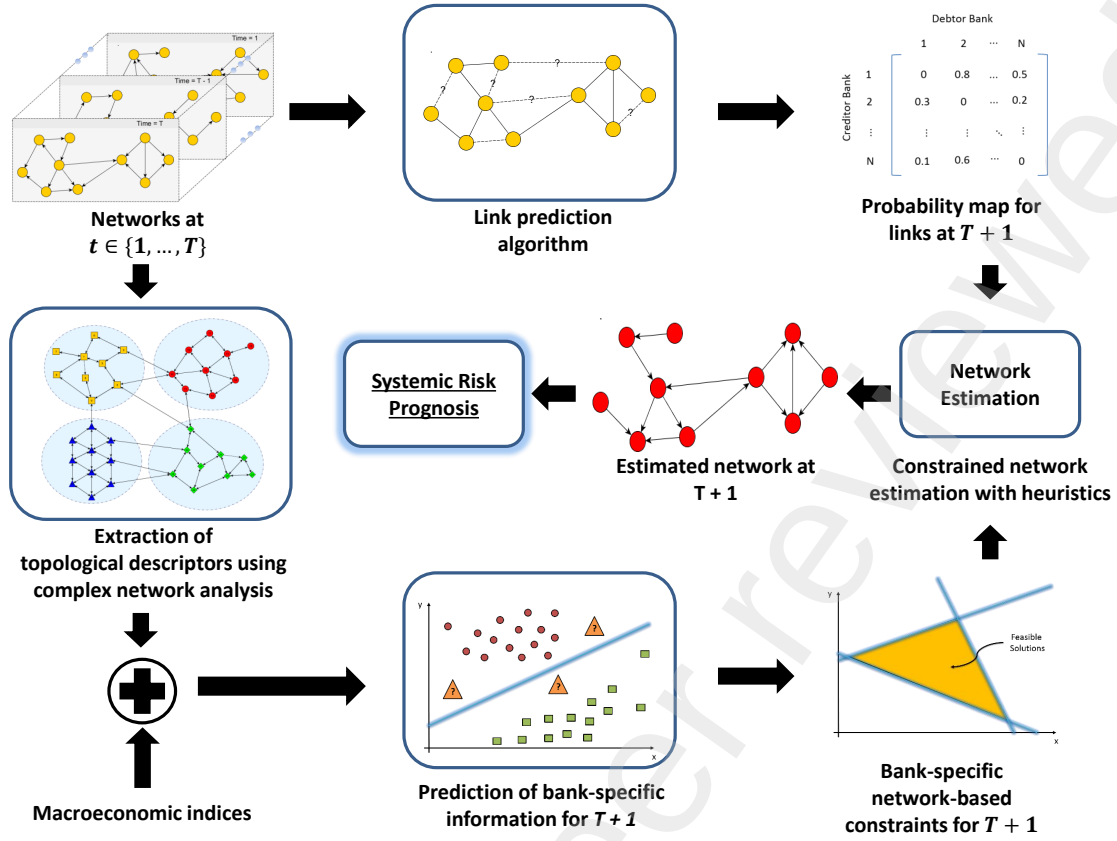


Figure 1: Overall scheme of the proposed framework for systemic risk prognosis. The steps surrounded by a hollow rounded rectangle are points of extension and are the main actors in the proposed scheme, as they concentrate on complex data processing methods. Steps that do not have rectangles are inputs or outputs for data processing steps.

of spurious links, we apply a cross-validation procedure to conduct the model selection using *only* data from the training set. To do so, we sub-divide the training set into two disjoint sub-sets: the *reduced training set* and a *validation set*. We use the reduced training set to construct our model, the link probability map, and the validation set to test the model's prediction power in a controlled environment.⁵ The model selection procedure is detailed in Section 3.

- **Prediction of network-based descriptors:** given the network snapshots ranging from $t \in \{1, \dots, T\}$, we extract several topological and structural features of these networks (the measurements are in Section 2.4). In addition, we consider macroeconomic indices to deal with unaccounted exogeneity or business cycles. We then use panel-data techniques to predict the structural features of the network at $t = T + 1$. We use these predicted values as constraints in the subsequent step of the network estimation. Specifically, we estimate the in- and out-strength of each bank, which gives us upper bounds for the lending and borrowing volumes of the bank at the predicted network in $T + 1$.
- **Network estimation process:** using the link probability map together with the estimated in- and out-

⁵For instance, we can select the networks from $t \in \{1, \dots, T - k\}$, $k \in \{1, \dots, T - 1\}$, as members of the reduced training set, and the network at $t = T - k + 1$ as the validation set. We use the reduced training set to train or calibrate the model, while the validation set is employed only to test the model's accuracy. Note that, in any case, in the model selection phase, we never predict $t = T + 1$, as it is an out-of-sample instance and no correct response is known. As such, we would never be able to calibrate our model.

strengths, we perform a constrained network estimation process guided by stochastic heuristics to construct the network at a posterior time $T + 1$. The constraints come from the in- and out-strengths and the heuristics from the link probability matrix. While the link probability matrix is good at providing the likelihood of the existence of links in the estimated network, it does not reasonably estimate the weight of such links. This is exactly where the predicted constraints of banks' in- and out-strengths fulfill their roles because they provide upper bounds for the total intra-network investments and funding for each bank. Here, we must also perform model selection to prevent trained models (estimated networks) that overfit or bias the underlying real networks. For that end, we also employ a cross-validation procedure with some interesting caveats (details are given in Section 3), as graph comparison is not a straightforward task to perform⁶.

- *Systemic risk prognosis*: with the estimated network at a posterior time $T + 1$ at hand, we are free to estimate several systemic risk measures proposed in the literature but in a forward-looking way. Regulators can use this information to verify in which direction the interbank network is heading, such as to a more risky network topology. If the regulator observes that the financial system is evolving to a more fragile state, it may anticipate the execution of policies aimed at correcting this distortion in a timely fashion and possibly in a more efficient manner than it would have happened had the regulator taken actions in a reactive way, rather than proactive.

2.3 Derivation of the probability matrix using link prediction

In this section, we show how the probability matrix that carries the likelihood of the existence of links between pairs of vertices is derived. This step is depicted in Fig. 1 as “Link prediction algorithm.” The network trajectory up to time T is symbolized here as $\mathcal{T}(T) = \{\mathbf{X}(1), \dots, \mathbf{X}(T)\}$. Once observed $\mathcal{T}(T)$, the link probability matrix for $t = T + 1$ is denoted here as $P^{(\text{link})}(\mathbf{X}(T + 1) | \mathcal{T}(T))$. The matrix $P^{(\text{link})}(\mathbf{X}(T + 1) | \mathcal{T}(T))$ is responsible for providing guidelines for the network estimation method in the link generation process, acting as a link distribution for $t = T + 1$.

2.3.1 Collapsing the network trajectory

We have a series of network snapshots throughout time $\mathcal{T}(T) = \{\mathbf{X}(1), \dots, \mathbf{X}(T)\}$, which together form a network tensor⁷. We must collapse this tensor into a resulting matrix using a pre-defined strategy to apply the link prediction algorithm. In this work, we follow the idea discussed in Dunlavy et al. (2011), in which the link structure is damped backward in time. We evaluate the collapsed network trajectory $\mathbf{Z}(T)$ as:

⁶In this paper, we discuss the equivalency of graph comparison to the classical graph isomorphism problem in graph theory, which is an NP-complete problem. Hence, algorithms of this kind cannot deal with large networks as they are exponential in their time computational complexity. We will solve this problem here by using efficient graph kernels.

⁷Tensors are matrices with 3 or more modes (dimensions). Each mode is a dimension of the matrix. In our network trajectory, there are three modes: creditor bank, debtor bank, and time. For a thorough review of tensors, the reader is referred to Boccaletti et al. (2014).

$$\mathbf{Z}(T) = \sum_{t=1}^T (1 - \theta)^{T-t} \mathbf{X}(T-t), \quad (2)$$

where $\theta \in [0, 1)$ represents the forgetting factor, or, conversely, $1 - \theta$ is the persistence factor. Observe that, as θ grows larger, less relevance is given for networks far behind the current time T . In contrast, when θ is small, there is a large persistence over time, and long-established links between banks are remembered. In the extreme, when $\theta = 0$, (2) reduces to:

$$\mathbf{Z}(T) = \sum_{t=1}^T \mathbf{X}^{(T-t)}, \quad (3)$$

i.e., the representative matrix $\mathbf{Z}(T)$ has infinite memory. Note that the (i, j) -entry of $\mathbf{Z}(T)$, $Z_{ij}(T)$, can be understood as an overall weighted and damped exposure from i to j over the studied period $t \in \{1, \dots, T\}$. As we will see in Section 3, a model selection procedure is performed to calibrate the forgetting factor θ , which plays an essential role in the link prediction algorithm.

2.3.2 Definition of the link prediction algorithm: truncated Katz index

We use the collapsed matrix $\mathbf{Z}(T)$ as input for the link prediction algorithm. This matrix embodies the temporal dynamics of the network in a summarized manner, for it penalizes distant observations in favor of more recent relationships whenever $\theta < 1$. In this paper, we opt to employ the truncated Katz index as for the link prediction method for two reasons: (a) the scores provided by the Katz indices are arguably one of the best link prediction schemes available in the literature as it has consistently outperformed several competing techniques (cf., for instance, Liben-Nowell and Kleinberg (2007)); (b) the truncation of the Katz index prevents matrix inversion evaluations, which are expensive, time-consuming procedures with a time computational complexity of the order $\mathcal{O}(N^3)$.⁸ First, we detail the computation of the Katz scores in an exact manner, followed by its truncated version. Afterward, the probability map for link prediction of the network at $T + 1$ is defined using the scores produced by the Katz indices.

Definition of the Katz index

Katz (1953) defines a measure that directly sums over the collection of paths in a graph, in which larger paths are exponentially damped according to their length. In this way, shorter paths count more heavily in the overall computation. The matrix that carries the Katz scores is given by:

$$\mathbf{S} = \sum_{l=1}^{\infty} \beta^l \mathbf{P}^{(l)} = \sum_{l=1}^{\infty} \beta^l \mathbf{Z}^l(T), \quad (4)$$

⁸The selection of the truncated Katz index confers scalability to our methodology and hence makes feasible its usage in banking systems with several banks.

where β is an attenuating or dampening factor. The matrix $\mathbf{P}^{(l)}$ stores the number of paths from all pairs of vertices in the network. For instance, $\mathbf{P}_{ij}^{(l)}$ is the number of paths with length l starting from $i \in \mathcal{N}$ and ending at $j \in \mathcal{N}$. The second equivalence in (4) comes from the fact that the number of paths of length l from i to j can be evaluated by raising to the power of l the representative matrix $\mathbf{Z}^{(T)}$, denoted here as $\mathbf{Z}^l(T)$. Observe that $\mathbf{Z}(T)$ is a weighted directed graph so that the paths are weighted according to the influence of each edge weight. This contrasts with the original definition of Katz (1953), who uses the adjacency matrix of the graph instead of the weighted matrix⁹. The (i, j) -entry of $\mathbf{S}$ in (4), S_{ij} , is the Katz score of the link starting from $i \in \mathcal{N}$ and terminating in $j \in \mathcal{N}$. We will see that this index can be understood as the likelihood of maintenance of that link at posterior periods.

Equation (4) only converges if $\beta \mathbf{Z}(T)$ has a spectral radius smaller than one. If $|\lambda_1|$ is the largest eigenvalue of $\mathbf{Z}(T)$ in magnitude, that condition is the same as to state that (4) converges when $\beta < 1/|\lambda_1|$. When the convergence constraint is not satisfied, $\mathbf{S}$ is not well-defined anymore, and, computationally, we verify the appearance of negative entries in the matrix representing the Katz indices. Now, if $\beta < 1/|\lambda_1|$, then the summation in (4) is a geometric series that converges, leading to the following matricial form:

$$\mathbf{S} = \sum_{l=1}^{\infty} \beta^l \mathbf{Z}^l(T) = (\mathbf{I} - \beta \mathbf{Z}(T))^{-1} - \mathbf{I}, \quad (5)$$

where $\mathbf{I}$ is an $N \times N$ identity matrix.

Definition of the truncated Katz index

Equation (5) is not scalable due to the matrix inversion operation that is of the order $\mathcal{O}(N^3)$, where N is the dimension of the square matrix. As such, computing the Katz scores in an exact manner is very time-consuming for large graphs. Given this shortcoming, Liben-Nowell and Kleinberg (2007) propose a low-rank approximation of $\mathbf{Z}(T)$ to compute the Katz scores more efficiently. Assume that $\mathbf{Z}$ has rank $R \leq N$, whose eigendecomposition is evaluated as:

$$\mathbf{Z}(T) = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T, \quad (6)$$

where the superscript T is the transpose operator, $\mathbf{V}$ is a $N \times N$ orthogonal matrix, whose columns are the eigenvectors ordered in non-increasing order according to the magnitude of their corresponding eigenvalues. The matrix $\mathbf{\Lambda}$ is a $N \times N$ diagonal matrix, whose main diagonal possesses the eigenvalues of $\mathbf{Z}(T)$ in non-increasing order, i.e., $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_R| > \lambda_{R+1} = \dots = \lambda_N = 0$. Rewriting (5) using the eigendecomposition shown in (6), we get:

⁹Recall (1) for obtaining the adjacency matrix from the weighted matrix.

$$\mathbf{S} = (\mathbf{I} - \beta \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T)^{-1} - \mathbf{I} = \mathbf{V} [(\mathbf{I} - \beta \mathbf{\Lambda})^{-1} - \mathbf{I}] \mathbf{V}^T = \mathbf{V} \mathbf{\Psi} \mathbf{V}^T, \quad (7)$$

where $\mathbf{\Psi} = (\mathbf{I} - \beta \mathbf{\Lambda})^{-1} - \mathbf{I}$. The second equivalence comes from the orthogonality of matrix $\mathbf{V}$, which is composed of the eigenvectors of $\mathbf{Z}(T)$ ¹⁰. Using the properties of matrix eigenvalues, we can link the eigenvalues of $\mathbf{\Psi}$ to the ones of $\mathbf{Z}(T)$ as follows:

$$\psi_i = \frac{1}{1 - \beta \lambda_i} - 1, \quad (8)$$

in which $\psi_i = 0, \forall i > R$ because $\mathbf{Z}(T)$ is a rank- R matrix by definition. Hence, we can drop all the rows and columns of $\mathbf{\Psi}$ whose indices are greater than R without loss of generality. In this way, we consider that $\mathbf{\Psi}$ is an $R \times R$ diagonal matrix with only the non-zero eigenvalues. Consequently, $\mathbf{V}$ transforms into an $N \times R$ orthogonal matrix, whose columns are the eigenvectors of the non-zero eigenvalues of $\mathbf{Z}(T)$ in non-decreasing order.

The best rank- L approximation of $\mathbf{Z}(T)$ is given by replacing $\mathbf{\Lambda}$ in (6) with a matrix $\mathbf{\Lambda}_L$ where only the L largest eigenvalues are maintained, i.e., $\mathbf{\Lambda}_L$ is of dimension $L \times L$. Using this fact and performing the same steps as done with the full matrix $\mathbf{Z}(T)$ with rank R , we end with a similar expression as (7), as follows:

$$\hat{\mathbf{S}} = \mathbf{V} \mathbf{\Lambda}_L \mathbf{V}^T, \quad (9)$$

where $\hat{\mathbf{S}}$ is an approximate version of the true Katz scores matrix $\mathbf{S}$ and $\mathbf{V}$ is a $N \times L$ matrix, where the columns correspond to the L largest eigenvalues of $\mathbf{Z}(T)$.

Liben-Nowell and Kleinberg (2007) show that the eigendecomposition of a sparse rank- L matrix can be efficiently evaluated using the Arnoldi or Lanczos method in $\mathcal{O}(\text{nnz}(\mathbf{Z}))$ per iteration, where $\text{nnz}(\mathbf{Z})$ is the number of non-zeros entries in $\mathbf{Z}$. Since interbank networks are usually very sparse, this stands as an efficient way of calculating the Katz scores.

Definition of the probability map

The link prediction algorithm of our framework accepts three input parameters: (a) the forgetting factor $\theta \in [0, 1)$, used to collapse the network history $\mathcal{T}(T)$; (b) the dampening factor β , used to guarantee the convergence of the Katz index computation, and (c) the scalar $L < R$, which corresponds to the L largest eigenvalues of the collapsed matrix $\mathbf{Z}$. Results strongly depend on the first input, while the second and third parameters are mathematical convenience tools.

¹⁰Recall that, if $\mathbf{M}$ is orthogonal, then $\mathbf{M}^T = \mathbf{M}^{-1}$, such that $\mathbf{M} \mathbf{M}^T = \mathbf{M} \mathbf{I} \mathbf{M}^T = \mathbf{I}$.

The output of the link prediction algorithm based on the Katz scores produces a link probability matrix or map for time $T + 1$, denoted here as $P^{(\text{link})}(\mathbf{X}(T + 1) \mid \mathcal{T}(T))$. Mathematically, it is given by:

$$P^{(\text{link})}(\mathbf{X}(T + 1) \mid \mathcal{T}(T)) = \frac{1}{k} \hat{\mathbf{S}}, \quad (10)$$

where k is a normalization constant that transforms the Katz score in a link distribution matrix and $\hat{\mathbf{S}}$ is the Katz score evaluated in accordance with (9). The normalization constant k is expressed as:

$$k = \sum_{i=1}^N \sum_{j=1}^N \hat{S}_{ij}. \quad (11)$$

Note that if banks i and j have several different paths connecting one to another, then S_{ij} is expected to be large, yielding a high probability value for the likelihood of existing a link between i and j at time $T + 1$, i.e., $P^{(\text{link})}(\mathbf{X}_{ij}(T + 1) \mid \mathcal{T}(T)) \rightarrow 1$. Conversely, when they are linked to each other with a small number of distinct paths, then S_{ij} is small, producing a low probability for the existence of a link between i and j . The logic here is that by compressing the network tensor into a single summarizing matrix, links that are persistent over time will be given a larger probability than those that are eventual links. This is the reason behind using the trajectory and not only the last snapshot of the network.

The link probability matrix $P^{(\text{link})}(\mathbf{X}(T + 1) \mid \mathcal{T}(T))$ is used as a heuristic for the underlying non-observable link generation process of the network at the posterior time $t = T + 1$. However, while this link probability matrix is a convenient mathematical tool for determining the existence or absence of links, it does not provide reasonable estimates of how big the weights linking two different arbitrary vertices are. To overcome this drawback, we will use bank-specific constraints extracted from the interbank network and from macroeconomic variables (cf. Sections 2.4 and 2.5).

2.4 Extraction of topological descriptors using complex network analysis

The link probability matrix $P^{(\text{link})}(\mathbf{X}(T + 1) \mid \mathcal{T}(T))$ measures the likelihood that links coexist in the network. However, it does not provide information on how strong the links connecting different vertices are. Since interbank networks are naturally represented by weighted directed networks, this feature is an important point in a network estimation process, as gross exposure values are vital for systemic risk analysis. To overcome that, we combine the link prediction routine with quantitative constraints that give upper bounds of the total investment (out-strength) and funding (in-strength) each bank can perform in the predicted network at $t = T + 1$. We use complex network theory and macroeconomic indices that embody economic conditions and business cycles to estimate these upper bounds.

We review some complex network indicators in the following, which can be of three types: strict-local, mixed, or global.¹¹

¹¹For a comprehensive review, refer to Silva and Zhao (2016).

2.4.1 Strict-local measures

Strict-local measures relate to the bank's characteristics. They do not take into account the features of the neighborhood or the network. We present some of these network measures in the following.

Degree

The degree or valency of a vertex $i \in \mathcal{N}$, indicated by k_i , is related to the connectivity, or number of links, of a vertex to the remainder of the network. In directed graphs, this notion can be further decomposed into the in-degree, k_i^{in} , and out-degree, k_i^{out} , such that the identity $k_i = k_i^{\text{in}} + k_i^{\text{out}}$ holds. The domain of k_i corresponds to the discrete-valued interval $\{0, \dots, V - 1\}$ if self-loops are not allowed. When $k_i = 0$, we say vertex i is a singleton vertex. Conversely, when k_i is sufficiently large, we say vertex i is a hub.

The in- and out-degree of vertex $i \in \mathcal{N}$ are defined as follows:

$$k_i^{(\text{in})} = \sum_{j \in \mathcal{N}} \mathbb{1}_{\{A_{ji} > 0\}}, \quad (12)$$

$$k_i^{(\text{out})} = \sum_{j \in \mathcal{N}} \mathbb{1}_{\{A_{ij} > 0\}}, \quad (13)$$

where $\mathbb{1}_{\{K\}}$ represents the Heaviside function that yields 1 if K , a logical expression, evaluates to true, and 0, otherwise. In a network of exposures, the in-degree represents the number of FIs in which participant i has invested (is exposed to). With a similar reasoning, the out-degree symbolizes the number of debtor FIs participant i has (they are exposed to i).

The in- and out-degree of a vertex can be understood as measures for quantifying the liabilities and assets diversification, respectively. As these indices grow, the more diversified an FI is in funding (liabilities side) or investments (assets side). In addition, FIs with large degrees are more susceptible to within-network events, as it is more likely that there will be paths connecting the origin of the event to the refereed FI. Among the several examples of possible events studied by the literature, we highlight the occurrence of idiosyncratic or joint defaults. However, the topological aspects of the network play a crucial role in the propagation of such events. As such, an FI with a large degree will not necessarily suffer a larger impact than an FI with a small degree, as the interbank connectivity pattern may amplify or reduce the impact accordingly to the direct or indirect neighborhoods of these FIs.

Strength

The strength of a vertex $i \in \mathcal{N}$, indicated by s_i , represents the total sum of weighted connections of i towards its neighbors. We can decompose the strength into the in-strength, s_i^{in} , and out-strength, s_i^{out} , such that the identity $s_i = s_i^{\text{in}} + s_i^{\text{out}}$ holds. The domain of s_i corresponds to the continuous interval $[0, \infty)$.

The in- and out-strength of vertex $i \in \mathcal{N}$ are defined as:

$$s_i^{(\text{in})} = \sum_{j \in \mathcal{N}} A_{ji}, \quad (14)$$

$$s_i^{(\text{out})} = \sum_{j \in \mathcal{N}} A_{ij}. \quad (15)$$

In a network of exposures, the out-strength represents the amount of money that a bank has invested in that market, providing a measure of total exposure. In contrast, the in-strength symbolizes the amount of money a bank has received from bank counterparties. Finally, it is worth noting that strength is a generalization of the degree measure for weighted networks.

2.4.2 Mixed measures

Mixed measures consider the neighborhood's structural or topological characteristics of a specific vertex to render information.

Average shortest path

The average shortest (geodesic) path of vertex $i \in \mathcal{N}$ is given by:

$$\langle p_i \rangle = \frac{1}{V-1} \sum_{j \in \mathcal{N} - \{i\}} p_{ij}, \quad (16)$$

where p_{ij} denotes the length of the shortest path from i to j . If there is a single graph component, the domain of such measure is $[0, V-1]$; otherwise, pairs of vertices that are not connected receive $p_{ij} = \infty$ (cf. da F. Costa et al. (2005) for details). In our analyses, we discard the small-sized components, such that the upper limit of $\langle p_i \rangle$ exists. If $\langle p_i \rangle$ is small, then vertex i is central in the topological sense to its counterparties because it can reach the majority of the other nodes with ease. Conversely, when $\langle p_i \rangle$ is large, we say that i is peripheral to the network, as it takes longer to propagate events emanating from it.

From an economic point of view, the measure p_i can be regarded as the average length of intermediation chains that start from the FI $i \in \mathcal{N}$.

Betweenness

The betweenness of vertex i , B_i , is a centrality measure that quantifies the fraction of shortest paths between any given pairs of vertices $(k, j) \in \mathcal{N}^2$, $k \neq j \neq i$, in a network in which i is one of the intermediate vertices in the geodesic path of k to j . It was originally proposed by Freeman (1977). Mathematically, we have:

$$B_i = \sum_{(k,j) \in \mathcal{N}^2, k \neq j \neq i} \frac{\sigma_{kj}(i)}{\sigma_{kj}}, \quad (17)$$

where σ_{kj} quantifies the number of shortest paths starting from k and ending in j and $\sigma_{kj}(i)$ denotes the number of shortest paths starting from k and ending in j such that i is an intermediate vertex, i.e., is a member of the geodesic path.

Criticality

The criticality of the FI $i \in \mathcal{N}$, C_i , quantifies the impact of the transfers performed by i to their counterparties' liquid assets. It is the sum of the vulnerabilities of its creditors regarding their exposures to the FI and is given by:

$$C_i = \sum_{j \in \mathcal{N}} V_{ji} = \sum_{j \in \mathcal{N}} \frac{A_{ij}}{E_j}, \quad (18)$$

where $V_{ji} = A_{ij}/E_j$ quantifies the vulnerability of j with respect to the funding dependence on i , E_j indicates the readily available resources or capital buffer of bank $j \in \mathcal{N}$. We can conceive the criticality as a quasi-local measure of transmission of liquidity shocks between FIs. Note that, in the spectrum of criticality, the local importance of an FI is not directly related to its size; rather, it is represented by its creditors' vulnerabilities, measured by their net exposures/capital buffer ratios.

Dominance

The dominance of vertex $i \in \mathcal{N}$, D_i , measures the relative importance of i on its neighbors' operations. The dominance of i as a lender and as a borrower is given by:

$$D_i^{(\text{lender})} = \sum_{j \in \mathcal{N}} \frac{A_{ij}}{s_j^{(\text{in})}}, \quad (19)$$

$$D_i^{(\text{borrower})} = \sum_{j \in \mathcal{N}} \frac{A_{ji}}{s_j^{(\text{out})}}, \quad (20)$$

i.e., the dominance evaluates the fraction of funding to be received by i 's neighborhood from i itself. If vertex i is dominant, then it will be responsible for a large fraction of the funding portfolio of its neighborhoods. Conversely, if it is not dominant, this fraction will be, in general, small, resulting in a low D_i . The removal of a dominant FI may cause a significant impact on its direct neighbors in terms of liquidity.

2.4.3 Global measures

Global measures make use of all the relationships contained in the network to derive information. We show some in the following.

Density

The network density D , also known as network connectivity, for a directed network is defined as:

$$D = \frac{E}{\binom{V}{2}} = \frac{2E}{V(V-1)}, \quad (21)$$

where V and E represent the total number of vertices and edges, respectively. The density assumes values in the interval $[0, 1]$. When $D = 0$, we say that $\mathcal{G}$ is an empty graph. Conversely, when $D = 1$, $\mathcal{G}$ is said to be a complete or maximal clique graph. Often in the literature, networks can also be classified as sparse when D assumes values near 0 and dense otherwise. As a rule of thumb, when the number of edges in the networks is of the order of the number of vertices, i.e., $E = \mathcal{O}(V)$, the network is considered sparse.

Assortativity

Assortativity is a network-level measure that, in a structural sense, quantifies the tendency of nodes to link with similar nodes in a network Newman (2002). The assortativity coefficient r is computed as the Pearson's correlation of degrees of nodes in each connected pair. Positive values of r indicate that the network's pairs of nodes have nodes in the endpoints with similar degrees, while negative values indicate endpoints with different degrees (Newman (2003)). In general $r \in [-1, 1]$. When $r = 1$, the network has perfect assortative mixing patterns, while it is completely disassortative in the case $r = -1$. Considering that i_u and k_u represent the origin and destination of the u -th edge of a non-empty graph, respectively, and that $l = \sum_{(i,j) \in \mathcal{N}^2} A_{ij}$ denotes the total edge weight of the adjacency matrix (in our context, it is the exposures matrix), the assortativity r is evaluated as follows:

$$r = \frac{l^{-1} \sum_{u \in E} i_u k_u - \left[\frac{l^{-1}}{2} \sum_{u \in E} (i_u + k_u) \right]^2}{\frac{l^{-1}}{2} \sum_{u \in E} (i_u^2 + k_u^2) - \left[\frac{l^{-1}}{2} \sum_{u \in E} (i_u + k_u) \right]^2}. \quad (22)$$

According to Silva and Zhao (2012, 2015), understanding the assortative mixing patterns in complex networks is important for interpreting node functionality and for analyzing the global properties of the networks' components. Concerning the interbank market network, assortativity is important for the following reasons:

- When the network indicates high disassortative mixing, i.e., $r = -1$, most banks connect with other banks with dissimilar degrees. Considering that in these networks, only a few banks have several

connections (money centers), the onset of default in these money centers directly affects a large proportion of the network's vertices. In this configuration, all its neighbors that are vulnerable may, in turn, default, leading to the beginning of a contagion process throughout the network. This is true because the neighbors are likely to have a low diversification, making them highly dependent on the invested banks. Conversely, if a bank with a low degree defaults because it is probably connected to money centers, the chaining effect of subsequent defaults may be stopped by the money center. This is because it has high diversification and is thus probably not vulnerable to low-degree banks.

- When the network indicates high assortative mixing, i.e., $r = 1$, we expect that two types of clusters or communities will emerge in the interbank network: communities of high- and low-degree vertices. In each community, there will be a high density of edges connecting each pair of banks of the same group, while a sparse number of edges will exist that cross different communities. We expect that the network will not present money centers in the sense that there are a few highly connected FIs. The high-degree community must be a large group, or else its members would have a significant number of edges connecting them to FIs from the low-degree community. Concerning the low-degree community, because there is only a small quantity of edges per vertex, it is expected that the network's structure will present long linear chains of banks (large network diameter), each of which will, therefore, exhibit very low diversification. If a member of the high-degree community defaults, it will affect all of the connected high-degree neighbors and also a few low-degree vertices, which are more likely to be vulnerable given their lower diversification. Additionally, because the low-degree community has long linear paths of less-diversified FIs, if they are vulnerable (have low capital buffers), there is a higher probability that the process will cascade through the neighbors in a domino effect, onsetting a process of contagion with long paths. On the other hand, if a default occurs in a member of the low-degree community, then it is very unlikely that it will propagate to the money center cluster. However, depending on the FI's vulnerability, the same domino effect may occur in the low-degree community.
- When the network does not indicate edge attachment preferences globally, we expect the graph will have a slight mixture of assortative and disassortative trends in its various regions. During the calculation of the assortativity, these terms are canceled out, resulting in $r \approx 0$. In this configuration, we do not expect to encounter communities in the strong sense, as in the previous case.

2.5 Prediction of bank-specific aggregated network-based measurements

In this section, we show the econometric model we use to estimate the in- and out-strength at $T + 1$ given data ranging up to time T . This step is depicted as "Prediction of bank-specific aggregate information for $T + 1$ " in Fig. 1. Since investment and funding tend to be persistent at the aggregate level, we use the dynamic panel-data estimator of Blundell and Bond (1998) to capture the persistence of the dependent variables.

We run the following dynamic panel-data specification:

$$y_{i,t} = \alpha + \rho y_{i,t-1} + \beta \text{Network}_{i,t-1} + \gamma \text{Macroeconomic}_{t-1} + \varepsilon_{i,t} \quad (23)$$

in which $y_{i,t}$ is the dependent variable that can either be the in- or the out-strength; α is a constant, the vector $\text{Network}_{i,t-1}$ contains network measures that capture structural aspects of the role that bank i plays in the interbank market; the vector $\text{Macroeconomic}_{t-1}$ has macroeconomic indices, such as inflation, policy interest rates, unemployment, exchange rates, that capture time-varying homogeneous effects; and $\varepsilon_{i,t}$ is the residual.

While we only predict the in- and out-strength (dependent variables), we use all lagged network indicators as explanatory variables in the $\text{Network}_{i,t-1}$ vector: local (degree), mixed (average shortest path, betweenness, criticality, dominance), global (density, assortativity).

2.6 Constrained network reconstruction with heuristics

In this section, we detail how $\mathbf{X}(T+1)$ is estimated via a constrained network reconstruction process. This phase is represented by the step “Constrained network estimation with heuristics,” as shown in Fig. 1. We use the superscript $[k]$ to represent the actual state of a variable after $k \in \mathbb{N}$ iterations of our network estimation algorithm. For instance, $\mathbf{X}^{[k]}(T+1)$ denotes the estimate of $\mathbf{X}(T+1)$ after k iterations.

It is worth noting that our objective function in the optimization process has been inspired by the work proposed by (Anand et al. (2015)). However, several adaptations and extensions have been performed on that original technique, such as to accommodate our requirements for systemic risk prognosis. The network estimation procedure takes as inputs the link probability matrix, $P^{(\text{link})}(\mathbf{X}(T+1) \mid \mathcal{T}(T))$ (cf. Section 2.3,) and the posterior in- and out-strength of each participant bank, $s_i^{(\text{in})}(T+1)$ and $s_i^{(\text{out})}(T+1)$, respectively (cf. Section 2.5.)

The essence of the network reconstruction scheme resides in combining theoretical information arguments with economic incentives to produce networks preserving realistic characteristics. One key point in the spectrum of interbank networks is that several works, among which we may highlight Boss et al. (2004); Castro Miranda et al. (2014), have shown that interbank networks are sparse, for interbank activity is based on costly relationships. In an optimization process, this sparseness property is consistent with the design of an objective function that attempts to minimize the number of necessary links for transmitting a specific volume of interbank loans. Following the terminology suggested by Anand et al. (2015), these network estimation techniques can be best described as “minimum density” (MD) approaches, in which the most probable links are identified and loaded with feasible exposures that respect constraints imposed on the optimization process.

Definition of the objective function

In the MD approach, consider that $c > 0$ represents the fixed cost of establishing a link. We aim to estimate $\mathbf{X}(T+1)$. The objective or the energy function of the maximization problem is formulated as:

$$\max_{\mathbf{X}^{[k]}(T+1)} -c \sum_{i=1}^N \sum_{j=1}^N \mathbb{1}_{\{\mathbf{X}_{ij}^{[k]}(T+1) > 0\}}, \quad (24)$$

subject to:

$$\begin{aligned} \sum_{j=1}^N \mathbf{X}_{ij}^{[k]}(T+1) &= s_i^{(\text{out})}(T+1), & \forall i \in \mathcal{N}, \\ \sum_{i=1}^N \mathbf{X}_{ij}^{[k]}(T+1) &= s_j^{(\text{in})}(T+1), & \forall j \in \mathcal{N}, \\ \mathbf{X}_{ij}^{[k]}(T+1) &\geq 0, & \forall (i, j) \in \mathcal{N}^2, \end{aligned} \quad (25)$$

where the operator $\mathbb{1}_{\{\cdot\}}$ denotes the Heaviside or indicator function, which yields 1 if its argument yields true, and 0, otherwise. $s_i^{(\text{out})}(T+1)$ and $s_i^{(\text{in})}(T+1)$ represent the total intra-network assets and liabilities of the i -th bank at $T+1$, here also called marginals, which are quantified according to the predicted out- and in-strength values by the algorithm explained in Section 2.5.

Despite being formally correct, the maximization problem previously defined has a very restrictive set of feasible solutions and is prone to numerical instability during the computational optimization task. As a way to alleviate this issue and still achieve solutions that fit well to the total assets and liabilities marginals, we follow the suggestion of Anand et al. (2015), in which these restrictions are softened by allowing deviations from the marginals that are penalized proportionally to their magnitudes. With this modification, the problem can be redefined as a maximization task of the objective function $V^{[k]}$, whose analytical form is:

$$V^{[k]} = -c \sum_{i=1}^N \sum_{j=1}^N \mathbb{1}_{\{\mathbf{X}_{ij}^{[k]}(T+1) > 0\}} - \sum_{i=1}^N \left(\alpha_i (AD_i^{[k]})^2 + \delta_i (LD_i^{[k]})^2 \right), \quad (26)$$

where $AD_i^{[k]}$ and $LD_i^{[k]}$ are given by:

$$\begin{aligned} AD_i^{[k]} &= s_i^{(\text{out})}(T+1) - \sum_{j=1}^N \mathbf{X}_{ij}^{[k]}, \\ LD_i^{[k]} &= s_i^{(\text{in})}(T+1) - \sum_{j=1}^N \mathbf{X}_{ji}^{[k]}. \end{aligned} \quad (27)$$

i.e., they are the current deviations at iteration k of the i -th bank from its pre-defined $s_i^{(\text{out})}(T+1)$ and $s_i^{(\text{in})}(T+1)$, which were estimated in the previous step of our framework. The exogenous parameters α_i and δ_i in (26) calibrate the penalty that is given for each unity that the current values deviate from their predicted values.

Initial condition setup

The optimization problem requires a suitable initialization of $\mathbf{X}(T+1)$ to run. Algorithmically, we need to set the states of all variables at iteration $k=0$. Instead of using a random or even a zero-valued matrix, we take as an initial guess the snapshot of the most recent known network $\mathbf{X}(T)$, i.e.:

$$\mathbf{X}^{[0]}(T+1) = \tau \mathbf{X}(T) \quad (28)$$

where the constant $\tau \in [0, 1]$ is a penalizing factor that prevents the algorithm from terminating if the network does not change much regarding the assets and liabilities marginals.

If we choose a large τ , then we are implicitly assuming that the estimated network at $t = T+1$ will not change much from its initial guess. This only holds if the considered period is not long. In periods of turmoil, small-valued τ values are indicated, as the network's topological structure may significantly modify from one period to another. In this scenario, this argument still may hold true even for small intervals.

Heuristics for finding approximate optimal solutions in the search space

The search space of the optimization problem defined by (26) and (27) comprises all possible matrices $\mathbf{X}$ with dimensions $N \times N$, whose entries are non-negative. As such, the computational complexity for navigating in this space is, in the binary case, $\mathcal{O}(2^{N^2})$ ¹². Consequently, the problem becomes computationally unfeasible even for small values of N . As such, heuristics for navigating in this search space must be designed.

Due to the initialization of $\mathbf{X}(T+1)$ in (28), some banks may lose or gain new edges (relationships) in between the previous and current iterations. In this way, we must define strategies for deleting, adding, and adjusting links in the network. Consider that $\mathbf{Q}^{[k]}$ is the probability matrix that our network estimation procedure is going to use at the iteration k . Then:

- If $\sum_{i \in \mathcal{N}} AD_i^{[k]} \leq 0$, then the current estimation of the network at iteration k , $\mathbf{X}^{[k]}(T+1)$, is still, in general, not completely built. In this case, our network reconstruction process is biased towards adding links until the network is completely structured, according to our predicted values of $s^{(\text{in})}(T+1)$. A perfect candidate for building the remainder of the network is our link probability matrix $P^{(\text{link})}(\mathbf{X}(T+1) \mid \mathcal{T}(T))$ defined in Section 2.3. As such,

$$\mathbf{Q}^{[k]} = P^{(\text{link})}(\mathbf{X}(T+1) \mid \mathcal{T}(T)). \quad (29)$$

¹²The binary adjacency matrix is a particular case and serves as a lower bound for the unrestricted optimization process. If the adjacency matrix is weighted, the base of the exponential changes accordingly. In any case, the optimization problem is computationally non-tractable (NP-complete) and scales exponentially to the number of banks (N).

- If $\sum_{i \in \mathcal{N}} AD_i^{[k]} > 0$, this means that the current estimated $\mathbf{X}^{[k]}(T+1)$ has assets marginals that are, in general, superior to the ones predicted by our regressions of $s^{(\text{in})}(T+1)$. In this situation, our network estimation model is biased towards deleting links. In this case, the network estimation algorithm employs the following probability matrix at iteration k :

$$\mathbf{Q}^{[k]} \propto \mathbf{A}^{[k]}(T+1) \circ \mathbf{D}^{[k]} \circ P^{(\text{remove})}(\mathbf{X}(T+1) \mid \mathcal{T}(T)), \quad (30)$$

where $\circ$ is the Hadamard product¹³.

The matrix $\mathbf{A}^{[k]}(T+1)$ in (30) is the adjacency matrix of the current estimate of $\mathbf{X}^{[k]}(T+1)$, which is constructed using (1). The introduction of $\mathbf{A}^{[k]}(T+1)$ is to prevent the selection of links that still do not exist in the estimated $\mathbf{X}^{[k]}(T+1)$, for we force a zero-valued probability to them.

The matrix $\mathbf{D}^{[k]}$ in (30) is given by:

$$D_{ij}^{[k]} = \left| \min \left\{ \frac{AD_i^{[k]}}{s_i^{(\text{out})}(T+1)}, -\varepsilon \right\} \times \min \left\{ \frac{LD_j^{[k]}}{s_i^{(\text{in})}(T+1)}, -\varepsilon \right\} \right|, \quad (31)$$

where $\varepsilon \ll 1$ is a sufficient small positive constant. The reasoning behind (31) is that we favor removing links that force banks to surpass their predicted out- and in-strength by a large amount. For instance, if the assets of the i -th bank are way too superior to its predicted value, then $AD_i^{[k]} < 0$, in light of (27). The same reasoning applies to the j -th bank that has current liabilities overestimated, resulting in $LD_j^{[k]} < 0$. If both deviations are negative, $D_{ij}^{[k]}$ is large, yielding a high probability of deletion of the link connecting i and j , because $Q_{ij}^{[k]}$ will be large. Now, when any of these deviations is not overestimated, then $D_{ij}^{[k]}$ will be small, resulting in a small-valued $Q_{ij}^{[k]}$. We have used a negative threshold $-\varepsilon$ instead of zero in the $\min\{\cdot\}$ functions in (31) to still be able to output non-zero probabilities for links in which just one of the sides is overestimated while the other is not. Now, if both sides are not overestimating, then $D_{ij}^{[k]} = \varepsilon^2$, which is a negligible quantity.

The matrix $P_L^{(\text{remove})}(\mathbf{X}(T+1) \mid \mathcal{T}(T))$ in (30) is derived based on our link probability matrix $P^{(\text{link})}(\mathbf{X}(T+1) \mid \mathcal{T}(T))$ explained in Section 2.3. Mathematically,

$$P_{ij}^{(\text{remove})}(\mathbf{X}(T+1) \mid \mathcal{T}(T)) \propto \begin{cases} \frac{1}{P_{ij}^{(\text{link})}(\mathbf{X}(T+1) \mid \mathcal{T}(T))}, & \text{if } P_{ij}^{(\text{link})}(\mathbf{X}(T+1) \mid \mathcal{T}(T)) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (32)$$

i.e., the probability matrix of removing links takes the reciprocal form of the probability of adding links, provided that we have positive entries in $P_{ij}^{(\text{link})}(\mathbf{X}(T+1) \mid \mathcal{T}(T))$. If $P_{ij}^{(\text{link})}(\mathbf{X}(T+1) \mid \mathcal{T}(T)) = 0$, then we do not need to output non-zero probabilities for removing them, as they will

¹³Also known as Schur or entry-wise product.

never exist by construction. Equation (32) is consistent with the fact that links with a small likelihood of existing should be removed with a high probability, such as to maintain the distribution found by the link prediction algorithm.

Acceptance criteria for modifications in the estimated network

Suppose that, at iteration k , the edge linking bank i to j is selected to be performed following the heuristics previously described. The algorithm may perform two actions: add or remove the current link. If the network estimation process chooses to add the link from i to j . In this case, the edge is incremented with a weight of:

$$\mathbf{X}_{ij}^{[k]}(T+1) = \mathbf{X}_{ij}^{[k-1]}(T+1) + \lambda_t \min\{AD_i^{[k]}, LD_j^{[k]}\}, \quad (33)$$

where $\lambda_t \in (0, 1]$ is the load factor. Observe that λ_t can be used to calibrate the density (connectivity) of the estimated network: as $\lambda_t \rightarrow 1$, the network becomes sparser. This point will be further studied in this work. In the model, the load factor is an annealing parameter as it increases over time by a factor $\Delta\lambda$. This procedure is performed to fasten the convergence of the algorithm.

The other decision that the network estimation algorithm can perform is to opt to remove the link from i to j . In this case, the algorithm destroys the link by setting:

$$\mathbf{X}_{ij}^{[k]}(T+1) = 0. \quad (34)$$

As such, negative adjustments can only be performed by first destroying the link and, afterward, by incrementing it several times. Due to the adapting bias inserted into the heuristics, this naturally occurs if we surpass the estimated values of the assets or liabilities marginals of a pair of banks.

After the selected action is performed by the network estimation algorithm, the objective function must be re-evaluated. Two different scenarios can occur:

- The action is always accepted if the objective function $V^{[k]}$ increases; and
- The action is accepted with a certain probability $P^{(\text{acceptance})}$ if the objective function $V^{[k]}$ decreases. If it is not accepted, then we re-run iteration k .

Suppose that $V^{[k]}$ decreased from one iteration to another by an amount of $\Delta V^{[k]} = V^{[k]} - V^{[k-1]}$. Then, the action is accepted with probability:

$$P^{(\text{acceptance})} = \exp\left(-\frac{|\Delta V^{[k]}|}{\max(\kappa_0 - \kappa k, 0)}\right), \quad (35)$$

where t is the iteration of the algorithm, κ_0 is the initial temperature of the simulated annealing process, and $\kappa \in (0, \infty)$ is the rate of adjustment of the temperature as the iterations progress. Note that the $P^{(\text{acceptance})}$ gets smaller as t increases because the current temperature of the estimation process, $\max(\kappa_0 - \kappa k, 0)$, decreases as k grows. For sufficient large k values, $P^{(\text{acceptance})} \rightarrow 0$. With this behavior, at the initial steps of the network formation method, we are more likely to accept links that decrease the objective function $V^{[k]}$. However, this probability diminishes as the temperature gradually increases in the process. This method is frequently used in stochastic estimation processes to prevent algorithms from getting trapped in local minima right out from the start due to the sloppy initial conditions.

Algorithm

Our network estimation algorithm is an iterative stochastic process. As such, we need to run it independently several times to obtain consistent network estimates. Algorithm 1 illustrates the main steps that compose our proposed network formation method for a single draw. Generally, it is advisable to run Algorithm 1 and choose the best-estimated network according to some quality criterion.

The algorithm accepts five input parameters: the network trajectory $\mathcal{T}(T)$ (denoted as $\mathcal{T}$); the link probability matrix $P^{(\text{link})}(X(T+1) | \mathcal{T}(T))$ (symbolized as $P^{(\text{link})}$); the posterior estimated in- and out-strengths $s^{(\text{in})}$, $s^{(\text{out})}$, respectively; the load factor λ , and the cost of maintaining a link c . Other less relevant parameters are not listed as input as they do not change the algorithm's output significantly. From lines 11 to 16, we force deletions as another mechanism for preventing local minima. These types of operations are called jumps in the search space and occur with a rare frequency of $\epsilon^{(\text{jump})} \ll 1$. In the line 42, $\epsilon^{(\text{termination})} \ll 1$, which is a termination criterion. Many termination criteria can be used, among which we may highlight when a pre-determined number of iterations k_{max} is reached or when the assets and liabilities marginal deviations reach a threshold.

2.7 Systemic risk prognosis

In this section, we show the systemic risk prognosis can be conducted. This step is represented in Fig. 1 as “Systemic Risk Prognosis.” At this step, the estimated posterior network is assumed as given since it is an output of the network estimation process. Network-based measures, besides the network structure, require the capital buffer of each bank to evaluate to what extent each bank can withstand losses. We can estimate the capital buffer using the same econometric setup as in Sect. 2.5.

With the predicted network structure and capital buffers, we can then virtually evaluate any network-based systemic risk indicator in the literature, such as the Eisenberg and Noe (2001)'s fictitious default procedure, Battiston et al. (2012a)'s DebtRank, Silva et al. (2017b)'s impact susceptibility and diffusion, among others.

Algorithm 1 Pseudo-algorithm for the constrained network formation method with heuristics.

```
1: procedure NETWORKESTIMATION( $\mathcal{T}, \mathbf{P}^{(\text{link})}, s^{(\text{in})}, s^{(\text{out})}, \lambda, c$ )
2:  $\mathbf{X}^{[0]} \leftarrow \text{SETINITIALNETWORKGUESS}(\mathcal{T})$  : Use (28)
3:  $\mathcal{L}^{(\text{present})} \leftarrow \text{GETPRESENTLINKS}(\mathbf{X}^{[0]})$ 
4:  $\text{AD}^{[0]} \leftarrow \text{CALCULATEASSETSDEVIATION}(\mathbf{X}^{[0]}, s^{(\text{out})})$  : Use (27)
5:  $\text{LD}^{[0]} \leftarrow \text{CALCULATELIABILITIESDEVIATION}(\mathbf{X}^{[0]}, s^{(\text{in})})$  : Use (27)
6:  $V^{[0]} \leftarrow \text{CALCULATEOBJECTIVEFUNCTION}(\mathbf{X}^{[0]}, c)$  : Use (26)
7:  $k \leftarrow 0$ 
8: repeat
9:    $k \leftarrow k + 1$ 
10:   $\rho \leftarrow \rho \sim U(0, 1)$ 
11:  if  $\rho < \varepsilon^{(\text{jump})}$  then
12:     $l_{ij} \leftarrow l_{ij} \sim U(\mathcal{L}^{(\text{present})})$ 
13:     $\mathbf{X}_{ij}^{[k]} \leftarrow 0$ 
14:     $\text{AD}_i^{[k]} \leftarrow \text{AD}_i^{[k-1]} + \mathbf{X}_{ij}^{[k-1]}$ 
15:     $\text{LD}_j^{[k]} \leftarrow \text{LD}_j^{[k-1]} + \mathbf{X}_{ij}^{[k-1]}$ 
16:     $\mathcal{L}^{(\text{present})} \leftarrow \mathcal{L}^{(\text{present})} - \{l_{ij}\}$ 
17:  else
18:     $[\mathbf{X}', \text{AD}', \text{LD}', \mathcal{L}'^{(\text{present})}] \leftarrow [\mathbf{X}^{[k-1]}, \text{AD}^{[k-1]}, \text{LD}^{[k-1]}, \mathcal{L}^{(\text{present})}]$ 
19:     $\text{bias} \leftarrow \text{EVALUATEBIAS}(\mathbf{X}', \text{AD}', \text{LD}')$ 
20:    if  $\text{bias} = \text{addLinks}$  then
21:       $\mathbf{Q} \leftarrow \mathbf{P}^{(\text{link})}$ 
22:       $l_{ij} = l_{ij} \sim \mathbf{Q}$ 
23:       $\mathbf{X}_{ij}' \leftarrow \text{SETNEWEDGEWEIGHT}(\mathbf{X}_{ij}', \lambda, \text{AD}'_i, \text{LD}'_j)$  : Use (33)
24:       $\text{AD}'_i \leftarrow \text{AD}'_i - \Delta \mathbf{X}_{ij}'$ 
25:       $\text{LD}'_j \leftarrow \text{LD}'_j - \Delta \mathbf{X}_{ij}'$ 
26:       $\mathcal{L}'^{(\text{present})} \leftarrow \mathcal{L}'^{(\text{present})} + \{l_{ij}\}$ 
27:    else
28:       $\mathbf{Q} \leftarrow \text{SETREMOVALPROBMATRIX}(\mathbf{X}', \mathbf{P}^{(\text{link})}, \text{AD}', \text{LD}', s^{(\text{in})}, s^{(\text{out})})$  : Use (30)
29:       $l_{ij} = l_{ij} \sim \mathbf{Q}$ 
30:       $\text{AD}'_i \leftarrow \text{AD}'_i + \mathbf{X}_{ij}'$ 
31:       $\text{LD}'_j \leftarrow \text{LD}'_j + \mathbf{X}_{ij}'$ 
32:       $\mathbf{X}_{ij}' \leftarrow 0$  : As (34)
33:       $\mathcal{L}'^{(\text{present})} \leftarrow \mathcal{L}'^{(\text{present})} - \{l_{ij}\}$ 
34:    end if
35:     $V^{[k]} \leftarrow \text{CALCULATEOBJECTIVEFUNCTION}(\mathbf{X}', c)$  : Use (26)
36:     $P^{(\text{acceptance})} \leftarrow \text{CALCULATEACCEPTANCEMATRIX}(\Delta V^{[k]})$  : Use (35)
37:     $\xi \leftarrow \xi \sim U(0, 1)$ 
38:    if  $V^{[k]} > V^{[k-1]}$  or  $\xi < P^{(\text{acceptance})}$  then
39:       $[\mathbf{X}^{[k]}, \text{AD}^{[k]}, \text{LD}^{[k]}, \mathcal{L}^{(\text{present})}] \leftarrow [\mathbf{X}', \text{AD}', \text{LD}', \mathcal{L}'^{(\text{present})}]$ 
40:    end if
41:  end if
42:  until  $\|\Delta V^{[k]}\| < \varepsilon^{(\text{termination})}$ 
43:  return  $\mathbf{X}^{[k]}$ 
44: end procedure
```

3 Model selection and prediction

The framework for systemic risk prognosis portrayed in Fig. 1 comprises several non-trivial steps in which computational models are designed. The models we construct in our framework are (a) link prediction model, (b) regression models for predicting network-based measurements and (c) network estimation model. In this section, we show how we can gauge how good our models are concerning some pre-defined criteria. Moreover, another complexity embodied in the proposed systemic risk prognosis framework is that it produces networks as outputs. To make the selection of the best model among several candidates viable, we must be able to compute how similar our generated networks are to the true ones. As we will see, comparing networks is not a straightforward task and will be tackled in this work by employing graph kernels. Once the model is calibrated correctly, we use it to predict unseen networks at posterior periods.

The outline of this part of the paper is as follows. Section 3.1 presents the well-known cross-validation methodology we use to prevent the construction of biased or overfitted models. Section 3.2 introduces the concept of graph kernels and how they can aid us in the graph comparison task. Section 3.3, we combine graph kernels with the cross-validation procedure to produce a robust and efficient model selection routine. Lastly, Section 3.4 provides how the forecasting mechanism is processed by the framework.

3.1 Cross-validation methodology

Cross-validation is a popular strategy for model selection. The main idea underlying the cross-validation procedure is to split the data, once or several times, in the training and the test sets. The part of the data in the training set is used to construct our model. Once the model is trained, we then use the data in the test set to measure its prediction power. The data from the training and the test sets must be disjoint; otherwise, we will create an optimistic bias. A fundamental assumption of cross-validation methods is that the data items are identically and independently distributed. The machine learning literature has proposed several methodological variations built on top of the traditional cross-validation model. Arlot and Celisse (2010) provide a comprehensive review of cross-validation methods and their variations for the interested reader. Some famous versions are the k -fold cross-validation and the leave-one-out. In the former, the data is divided into k folds, such that $k - 1$ folds are used for training and 1 folder for testing, which is employed for verifying the algorithm's performance. This configuration independently runs k times by shifting the test folder among the k possible folders. The leave-one-out is a special form of cross-validation in which each folder has only one element. Another particular variation of interest in cross-validation procedures is nested cross-validation. In this method, the training set is further subdivided into two other disjoint sets: the reduced training set and the validation set. Nested cross-validation is a commonly used approach to estimate the generalization performance of a modeling process, which includes internal model selection. It consists of two cross-validation procedures wrapped around each other. While the inner cross-validation is used for model selection, the outer one estimates generalization performance.

Recall, however, that we are dealing with a series of network snapshots taken at fixed intervals. In this way, these networks are expected, in some sense, to be correlated. As these data are not independent, traditional cross-validation methods cannot be used. Following the line of Bergmeir and Benítez (2012), we use a cross-validation procedure adjusted for dependent observations in time. In our case, assume that our training set is given by $\mathcal{R}_t^{(\text{train})} = \mathcal{T}(t) = \{\mathbf{X}(1), \dots, \mathbf{X}(t)\}$, and that the test set is represented by $\mathcal{R}_t^{(\text{test})} = \{\mathbf{X}(t+1)\}$, which is never used by the model selection routine. In this work, we apply the nested cross-validation by further dividing $\mathcal{R}_t^{(\text{train})}$ into $\mathcal{R}_t^{(\text{reducedTrain})} = \{\mathbf{X}(1), \dots, \mathbf{X}(t-1)\}$, and $\mathcal{R}_t^{(\text{validation})} = \mathbf{X}(t)$, $t \leq T$. The validation set $\mathcal{R}_t^{(\text{validation})}$ is used to select the best model according to some criteria. We can gauge the stability of our model chosen by selecting several $t' < T$, such that the test set becomes $\mathcal{R}_{t'}^{(\text{test})} = \{\mathbf{X}(t'+1)\}$, which is known, because $t' + 1 < T + 1$. If the model is stable, then its prediction power will not change much as we vary t' concerning some target measure, such as a stress index of systemic risk.

In a real-world estimation, we set $t = T$ because, in principle, we will make a prognosis for the

immediate unseen period $\mathbf{X}(T+1)$. In this configuration, we would use all of the network snapshots for the training phase because $\mathcal{R}_T^{(\text{train})} = \mathcal{T}(T) = \{\mathbf{X}(1), \dots, \mathbf{X}(T)\}$, and the test set would be $\mathcal{R}_T^{(\text{test})} = \{\mathbf{X}(T+1)\}$, as desired.

3.2 Graph comparison via graph kernels

In this section, we show how the graph comparison task is performed via efficient graph kernels.

Motivations for choosing graph kernels as a method for graph comparison

Although graphs render a great expressivity of representation by unifying the functions and dynamics of systems that they represent, this expressivity is achieved at a due price: many operations and metrics that are trivial for elementary data structures, such as the comparison of scalars, vectors, or matrices, become cumbersome and time-consuming for graphs. There is no known polynomial-time algorithm, even for the seemingly simple problem of deciding whether two graphs are equal. As Shervashidze et al. (2011) draws attention to, many of the most powerful machine learning techniques have their application limited to points in Euclidean spaces or require the ability to assess the similarity of points. For this reason, the mainstream research on graphs and machine learning tries to faithfully represent graphs in Euclidean or more general vector spaces so that those traditional operations are retrieved for graphs.

The graph comparison problem can be formally stated as follows. Given the set $\mathcal{G}$ of all possible graphs, the problem stands for finding a function $k: \mathcal{G} \times \mathcal{G} \mapsto \mathbb{R}$, such that $k(G_1, G_2), \forall G_1, G_2 \in \mathcal{G}$, quantifies the similarity of G_1 and G_2 . In graph theory, comparing graphs is considered a graph isomorphism problem. In this procedure, the graphs are considered isomorphic if they are topologically identical. This strategy gives rise to a binary similarity measure, which equals 1 if the graphs are isomorphic and 0 otherwise. Despite the idea of checking graph isomorphism being so intuitive, this task has been proven to be in the NP class of algorithms. However, it is still unknown whether graph isomorphism is NP-complete or can be solved by a polynomial-time algorithm (Garey and Johnson (1990)). Another natural approach to graph comparison is subgraph isomorphism, which is an analog of graph isomorphism but for graphs with different sizes. In contrast to the graph isomorphism problem, the subgraph isomorphism task has been proven NP-complete (Garey and Johnson (1990)). Even the more relaxed task of finding the largest common subgraph in two different graphs is NP-hard (Garey and Johnson (1990)). Given these particularities, the graph comparison becomes computationally intractable as the number of vertices increases.

Graph kernels have gained increasing attention from the community because they address the problems that traditional graph comparison approaches suffer, such as exponential runtime. These kernels are constructed to respect and exploit the graph topology but with some structural restrictions. Among them, the most common is the identification of substructures of graphs in a way that the kernel computation can still be evaluated in polynomial time.

Succinctly, a kernel is a bivariate function that quantifies how similar two input objects are. Mathematically, a kernel function corresponds to an inner product. Kernels are interesting because they can

supply a way of efficiently computing inner products in high-dimensional spaces, which is essential for many machine learning algorithms. The more similar the two objects, the larger the value yielded by kernels. Among the several kernels in the literature, the most famous ones are the Gaussian RBF and polynomial. Schölkopf et al. (2004) provide a comprehensive investigation on kernels.

Graph kernels are instances of the family of the so-called R -convolution kernels by Haussler (1999). R -convolution is a generic way of modeling kernels on discrete structured objects, such as graphs, by comparing all pairs of decompositions. In this way, each graph decomposition strategy results in a new graph kernel. However, the decomposition of a graph in arbitrary subgraphs is called the “all graphs” kernels that are at least as hard to compute as deciding if graphs are isomorphic (Shervashidze et al. (2011)). Therefore, in this work, we select graph kernels that compare only specific types of subgraphs to maintain their computation in polynomial runtime. Several different graph kernels have been defined in machine learning. They can be categorized into three classes: graph kernels based on walks and paths, graph kernels based on limited-size, and graph kernels based on subtree patterns. Some graph kernels accept labels in the vertex set, identifying the same vertex among different graphs. This information simplifies the process of graph comparison. In our case, since the participant banks in the network are known a priori, we opt to use these kinds of labeled graph kernels.

Weisfeiler-Lehman subtree graph kernel

In this work, we focus on an efficient kernel proposed by Shervashidze et al. (2011) called Weisfeiler-Lehman (WL) graph kernel, which is classified as a subtree kernel with labeling. This kernel is built upon a node-compressing algorithm called the Weisfeiler-Lehman algorithm. The key idea of this algorithm is to successively augment the node labels with the sorted set of node labels of the neighboring nodes and to compress these augmented labels into new, shorter labels. These steps are then repeated until the node label sets of the two compared graphs differ or until a maximum iteration is reached. Figure 2 portrays an overview of the WL algorithm for two arbitrary labeled graphs G_1 and G_2 .

Based on the WL relabeling algorithm, we formalize the WL subtree kernel in the following. Let $G_1 \in \mathcal{G}$ and $G_2 \in \mathcal{G}$ be arbitrary graphs. Let Σ be an alphabet containing all possible node labels. Say that $\Sigma^{[i]} \subseteq \Sigma$ is the actual alphabet contained in G_1 or G_2 at iteration i . Thus, $\Sigma^{[0]}$ is the set of original labels of G_1 and G_2 . Without loss of generality, assume that every $\Sigma^{[i]} = \{\sigma_{i,1}, \dots, \sigma_{i,|\Sigma^{[i]}|}\}$ is ordered¹⁴. Define the bivariate function $c_i : \mathcal{G} \times \Sigma^{[i]} \mapsto \mathbb{N}$, such that $c_i(G, \sigma_{i,j})$ is an indicator of the number of occurrences of the label $\sigma_{i,j} \in \Sigma^{[i]}$ in graph $G \in \mathcal{G}$.

The WL subtree kernel on graphs G_1 and G_2 with h iterations is defined as:

$$k_{WL}^{[h]}(G_1, G_2) = \langle \phi_{WL}^{[h]}(G_1), \phi_{WL}^{[h]}(G_2) \rangle, \quad (36)$$

where $\langle \cdot, \cdot \rangle$ represents the inner product of two quantities. $k_{WL}^{[h]}(G_1, G_2)$ is the WL subtree kernel value for graphs G_1 and G_2 at iteration h . Note that $k_{WL}^{[h]}(G_1, G_2)$ gets larger as G_1 and G_2 are more similar, due

¹⁴Recall that we can always map each entry of $\Sigma^{[i]}$ to a sequential identifier, such as by requiring ordering we do not lose generality. This holds because $\Sigma^{[i]}$ contains unique elements in a way that a bijective mapping always exists.

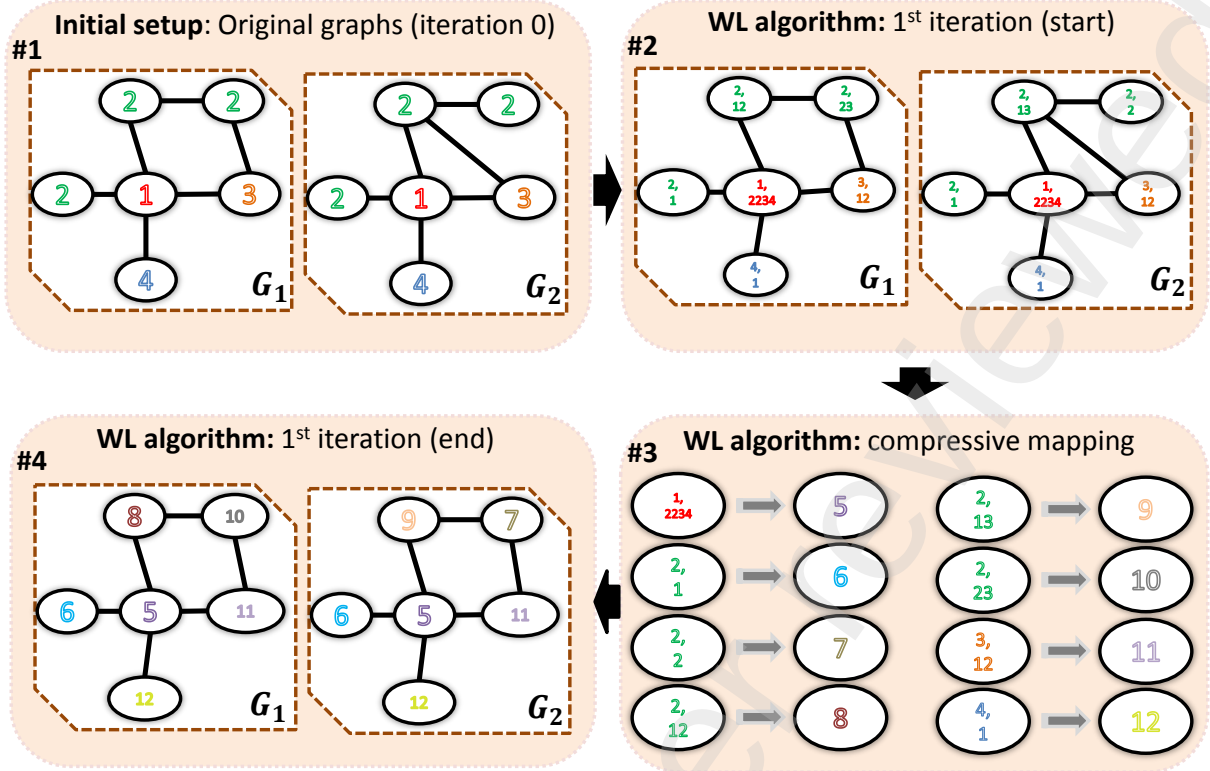


Figure 2: Simplified schematic of the steps involved in the Weisfeiler-Lehman algorithm. The numbers inside the nodes represent labels. Step #1 represents the initial setup, in which two graphs G_1 and G_2 are informed. Step #2 indicates the first label expansion to the immediate or direct neighborhood. Step #3 depicts the sequential label compressing strategy. Step #4 exhibits the final state of the first iteration of the WL algorithm, i.e., after the relabeling process.

to the inner product. The graphs are translated to feature vectors according to the nonlinear functions $\phi_{WL}^{[h]}$, located inside the inner product. This function lets graphs be comparable, and this is where the WL algorithm comes into play. The feature vector constructed from each graph corresponds to the frequency of all alphabets evaluated from iterations 0 to h in the WL algorithm exhibited in Fig. 2. This frequency is exactly supplied by the function c_i , $i \in \{0, \dots, h\}$. Mathematically, the nonlinear functions $\phi_{WL}^{[h]}$ have the form:

$$\phi_{WL}^{[h]}(G) = \left[\overbrace{c^{[0]}(G, \sigma_{0,1}), \dots, c^{[0]}(G, \sigma_{0,|\sigma_0|})}^{\phi_{WL}^{[h-1]}(G)}, \dots, c^{[h]}(G, \sigma_{h,1}), \dots, c^{[h]}(G, \sigma_{h,|\sigma_h|}) \right], \quad (37)$$

where $G \in \mathcal{G}$. If $h > 0$, then (37) can be further simplified by noting the recursive indirection:

$$\phi_{WL}^{[h]}(G) = [\phi_{WL}^{[h-1]}(G), c^{[h]}(G, \sigma_{h,1}), \dots, c^{[h]}(G, \sigma_{h,|\sigma_h|})]. \quad (38)$$

For the sake of completeness, the feature vectors extracted from the WL algorithm for G_1 and G_2 when $i = 0$ in Fig. 2 (Step #1) are:

$$\phi_{WL}^{[0]}(G_1) = [c^{[0]}(G_1, \sigma_{0,1}), \dots, c^{[0]}(G_1, \sigma_{0,4})] = [1, 3, 1, 4], \quad (39)$$

$$\phi_{WL}^{[0]}(G_2) = [c^{[0]}(G_2, \sigma_{0,1}), \dots, c^{[0]}(G_2, \sigma_{0,4})] = [1, 3, 1, 4]. \quad (40)$$

where $\Sigma^{[0]} = \{1, \dots, 4\}$. After the WL algorithm finishes the first iteration $i = 1$ in Fig. 2 (Steps #2 to #4), the feature vectors become:

$$\phi_{WL}^{[1]}(G_1) = [\phi_{WL}^{[0]}(G_1), c^{[1]}(G_1, \sigma_{1,5}), \dots, c^{[1]}(G_1, \sigma_{1,12})] = [\underbrace{1, 3, 1, 4}_{\phi_{WL}^{[0]}(G_1)}, 1, 1, 0, 1, 0, 1, 1, 1], \quad (41)$$

$$\phi_{WL}^{[1]}(G_2) = [\phi_{WL}^{[0]}(G_2), c^{[1]}(G_2, \sigma_{1,5}), \dots, c^{[1]}(G_2, \sigma_{1,12})] = [\underbrace{1, 3, 1, 4}_{\phi_{WL}^{[0]}(G_2)}, 1, 1, 1, 0, 1, 0, 1, 1]. \quad (42)$$

where $\Sigma^{[1]} = \{5, \dots, 12\}$ after the label compression is completed (cf. Step #3 in Fig. 2).

Now, with the feature vector representations of graphs G_1 and G_2 , we can then apply (36) to find the similarity between the two given graphs. For instance, if we select $h = 0$, the inner product between (39) and (40) must be performed, as follows:

$$k_{WL}^{[0]}(G_1, G_2) = \langle \phi_{WL}^{[0]}(G_1), \phi_{WL}^{[0]}(G_2) \rangle = 27, \quad (43)$$

and for $h = 1$, using (41) and (42), we get a kernel value of:

$$k_{WL}^{[1]}(G_1, G_2) = \langle \phi_{WL}^{[1]}(G_1), \phi_{WL}^{[1]}(G_2) \rangle = 27 + 3 = 30. \quad (44)$$

Note that $k_{WL}^{[h+1]}(G_1, G_2) \geq k_{WL}^{[h]}(G_1, G_2)$, because all of the entries in the graph feature vectors are non-negative. Now, if we want to test for graph isomorphism (strict equality), we must perform an element-wise comparison of the feature vectors from G_1 and G_2 . In this case, for $h = 0$, the graphs would be considered equal. In contrast, if $h = 1$, they would be interpreted as distinct, in that (41) and (42) are different.

Shervashidze et al. (2011) show that the WL subtree kernel on a pair of graphs G_1 and G_2 can be computed in time $\mathcal{O}(hE)$, where h is the maximum height of the subtree pattern and E is the number of edges. Taking into account that interbank networks are often very sparse, then the number of edges is of the order of the number of vertices, that is $\mathcal{O}(E) \approx \mathcal{O}(N)$. In this way, the overall time complexity of the WL subtree graph kernel is $\mathcal{O}(hV)$. Moreover, because $h \ll V$, h can be considered as a constant and, hence, can be dropped out of the order term. Thus, for usual interbank networks, the WL subtree graph kernel runs in roughly linear time $\mathcal{O}(V)$, which is a very welcomed property for large networks.

3.3 Model selection procedure

Grounded on the concepts of cross-validation and graph kernels presented in the previous sections, the model selection routine for the systemic risk prognosis framework is now discussed.

Overall strategy

Figure 3 shows how the model selection procedure is performed on the systemic risk prognosis framework. The framework's model selection comprises two independent inner model selection procedures: one based on pairwise individual information and one placed on aggregate information. The first one estimates a distribution that approximates the underlying network's link-generation process. The second one accounts for modulating the edge weights in the network formation step by giving upper bounds for the total investment and funding within-network for each bank. Note that we use the reduced training set ($t \in \{1, \dots, T-1\}$) to construct a model and use the validation set ($t = T$) to verify its performance. The performance assessment employs the WL subtree graph kernels that were previously introduced.

The reason behind devising the separation of the overall model selection into two independent blocks is given as follows. The model selection performed on the rectangle "pairwise individual information" in Fig. 3, which comprises the link prediction algorithm and the network formation technique, uses projected time-dependent network snapshots, following (2). As such, the temporal information or the actual network dynamics are not modeled by the link prediction and the network formation steps. Even though the summarization of the network snapshots retains some temporal information by reinforcing persistent links and weakening eventual links over different snapshots in time, these two routines assume a flattened data format. Thereby, they never truly explore the network dynamics.

The model selection performed on the "aggregate information" rectangle in Fig. 3 is more straightforward and uses classic econometric techniques to find the coefficients for $s^{(\text{in})}(T)$ and $s^{(\text{out})}(T)$, as ruled by (23).

The strategy for conducting the framework's model selection is as follows. First, we predict $s^{(\text{in})}(T)$ and $s^{(\text{out})}(T)$. Then, we jointly perform the model selection routine using the link prediction algorithm and the network formation technique using several parameter values. The combination of parameters that yield the lowest error is selected to go to the test phase, where we predict in an unseen network at $T+1$.

Model selection on the pairwise individual information

Table 1 supplies the parameters that must be calibrated in this phase. In Section 2.3, we have seen that link probability matrix estimation relies on a link prediction algorithm composed of two sequential phases. In the first phase, we summarize the network trajectory via a weighted projection of time-indexed network snapshots using (2). This network characterization depends on the forgetting factor θ . In the second phase, we construct the link probability matrix $P^{(\text{link})}(\mathbf{X}(T) \mid \mathcal{T}(T-1))$ using a truncated Katz

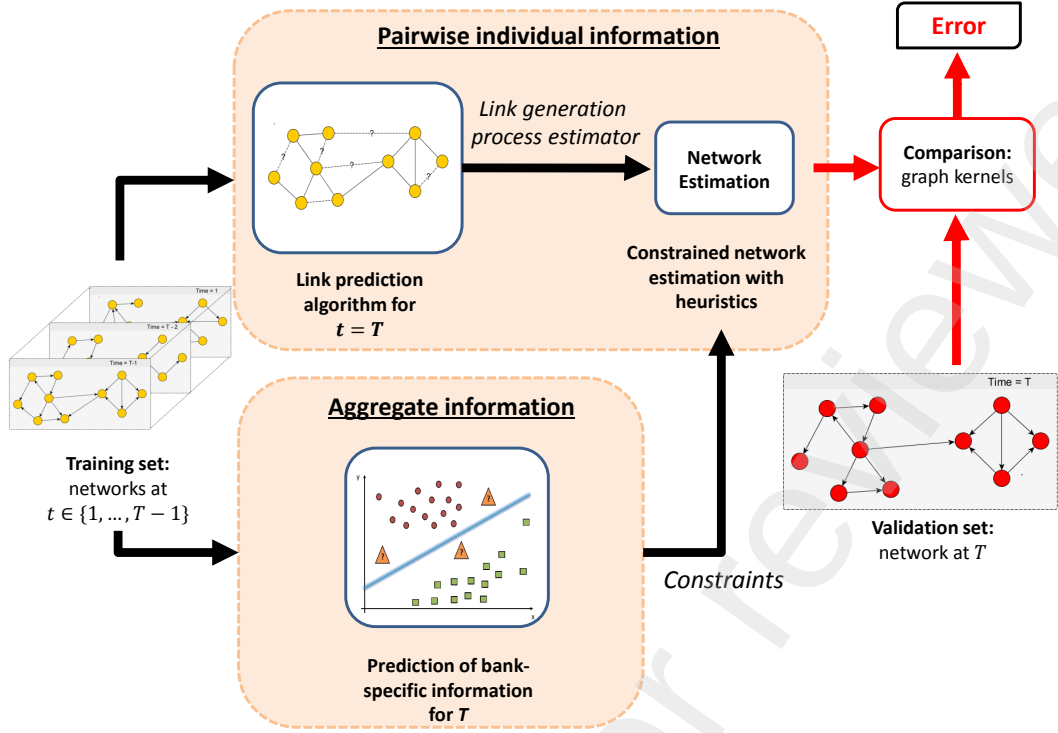


Figure 3: Schematic of the proposed model selection procedure for the systemic risk prognosis framework. The solid rectangular forms delineate where model selection is processed. Phases inside the same rectangular form symbolize that the selection of models is performed in a joint manner, not separately. The framework is divided into two main model selection procedures. One concerns pairwise individual information, and the other aggregates information. The computed model is tested against a validation set to evaluate its prediction power on unseen data.

scores¹⁵. In this phase, we must define β , which is the attenuating factor in (5), and L , which is the low-rank approximation of the exact Katz indices in (9). The model selection consists of properly calibrating θ , β , and L , as displayed in the first part of Table 1.

Provided that $\beta < 1/|\lambda_1|$, the parameter β is not sensitive to the outcome of the link prediction algorithm. Thus, the parameter β is a controllable feature as we know the largest eigenvalue of the collapsed weighted matrix once the forgetting factor is defined. Now for the two remaining parameters, we train the model using the following training set: $\mathcal{R}_T^{(\text{reducedTrain})} = \{\mathbf{X}(1), \dots, \mathbf{X}(T-1)\}$ and validate it against $\mathcal{R}_T^{(\text{validation})} = \{\mathbf{X}(T)\}$. In the model selection phase, the output of the link prediction algorithm will be the matrix containing the likelihood of links existing at $t = T$, i.e., $P^{(\text{link})}(\mathbf{X}(T) | \mathcal{T}(T-1))$.

In our model selection process, we search over the following discrete spaces. For the forgetting factor θ , we attempt values in $\{0, 0.05, \dots, 0.95\}$. For the number of low-rank components L , we search over $\{1, 2, \dots, R\}$, where R is the rank of the collapsed weighted matrix. If $L = R$, we end with the exact values for the Katz indices. In practice, when we are dealing with large-scale networks, we must fix a small L to be able to perform the model selection in a feasible time. Therefore, in some cases, the parameter L must be taken as given in the model selection procedure. The set of link probability matrices generated from the Cartesian combination of these parameters serves as inputs to the network estimation model.

¹⁵Remember that the model selection is always conducted using in-sample data (training set). The probability matrix $P^{(\text{link})}(\mathbf{X}(T+1) | \mathcal{T}(T))$, in turn, is estimated once the model selection is finished.

Table 1: Meta-parameters that must be calibrated by procedures in the pairwise individual information rectangle in Fig. 3.

Technique	Meta-parameter	Description
Link Prediction	1. Forgetting factor (θ)	Penalizer for distant network snapshots.
	2. Spectral radius modulator (β)	Assures that the Katz indices are bounded.
	3. Number of components to be used in the truncated Katz scores. (L)	Low-rank approximation of the Katz index.
Network Estimation	1. Link cost (c)	Cost of maintaining a relationship with another financial institution.
	2. Penalizer for deviations on total assets (α)	Lagrangian constraint for deviating from the total assets.
	3. Penalizer for deviations on total liabilities (δ)	Lagrangian constraint for deviating from the total liabilities.
	4. Initial load fraction (λ_0)	The fraction of the total possible value that is assigned to a link.
	4. Increment of the load fraction ($\Delta\lambda$)	The increment of the load fraction as time progresses.
	5. Link formation heuristics: (Q)	Heuristic matrix for selecting pairs of financial institutions.
	6. Link persistence attenuator: (τ)	Mirrors the last known snapshot of the network as the initial condition of the network estimation technique.

Concerning the network estimation model, our minimum density approach relies on an optimization process composed of three terms, as (26) reveals. In the second part of Table 1, we summarize all the parameters involved in the network estimation process. The penalizers for deviations on the constraints of total interbank assets and liabilities - α and δ , respectively - are equally important. As such, the meta-parameters α and δ must be similar. Moreover, the individual weighting factors c , α , and δ taken in an isolated form are irrelevant for the optimization process. The important factor is their relative value to each other. In this way, we can fix $\alpha_i = \delta_i = 1, \forall i \in \mathcal{N}$, with no loss of generality. As a result, the analysis of the cost weight (c) already encompasses the behaviors of the penalizers of the interbank total assets and liabilities. The velocity of the annealing parameter, $\Delta\lambda$, is deliberately set to a small value 10^{-6} , in such a way that the annealing process is very slow and the initial load fraction $\lambda = \lambda_0$ prevails during the iterations. We slow down the annealing mechanism to gauge the influence of the load fraction more precisely¹⁶. Had we left it with considerable values, we would have an effective λ in-between the region comprising the initial load fraction λ_0 and 1 (maximum possible value). This would bring unnecessary complexity to our investigation. As such, we are left to analyze the link cost parameter c and the initial, practically constant load fraction λ . With regard to the link formation heuristics Q , we use the output of the link prediction algorithm discussed above, i.e., the link probability matrix $P^{(\text{link})}(\mathbf{X}(T) | \mathcal{T}(T-1))$. The parameter τ is a hint for the network estimation technique of how likely are the last known network snapshot and the network to be estimated, using as an initial guess an attenuated view of the last known network snapshot weighted by τ (cf. (28)). Since the heuristics that are embodied in the network estimation procedure accept the construction and destruction of links, the parameter τ serves to speed up the

¹⁶We cannot turn off the annealing process due to convergence issues.

convergence of the algorithm. Due to that, we deliberately fix it at $\tau = 0.7$.

3.4 Prediction

Recall that the goal of the model selection procedure in Section 3.3 is to calibrate the link prediction and network estimation parameters, which are reported in Table 1. In the prediction phase, however, the goal is to predict for posterior periods. Applications that this forecasting can help are various. Among them, we highlight the possibility of regulators employing this estimated interbank network to gauge the systemic risk of the system in a proactive manner. This information, in turn, can be used to verify, for instance, if the interbank network is evolving to a more fragile state. For the prediction task to the immediate posterior period, $T + 1$, the test set is $\mathcal{R}_T^{(\text{test})} = \{\mathbf{X}(T + 1)\}$. In the prediction phase, we assume that all of the parameters displayed in Table 1 are fixed, according to our model selection procedure. That is, we consider that the models in the pairwise individual information rectangle in Fig. 3 are constants.

4 Conclusions

This paper contributes to the risk measurement debate in complex networks by proposing an innovative methodology for estimating complex networks. The methodology allows forward-looking approaches to assess future risks and, in particular, to measure systemic risk.

Instrumentally, the paper also innovatively tackles the nontrivial problem of comparing different complex networks. We use graph kernels that incorporate not only local but also global structural aspects of the network topology. Our approach departs from the traditional approach of comparing graphs using network descriptors, such as the average degree, strength, or centrality of network vertices, which are very weak because we can obtain totally different networks and still respect these constraints.

The methodology is sufficiently general to allow several potential applications. The estimated network may include interbank transactions between financial institutions, payments between financial institutions or companies, individuals, or derivative transactions.

This is the first paper that contributes to the estimation of future complex networks formed by economic agents in which links are estimated to guarantee certain economic rationality in their formation. As the method is sufficiently general, it can be used in combination with different risk and systemic risk measures to assess the financial fragility of financial systems or even as an instrument to gauge the financial stress of individual financial institutions.

References

- Acemoglu, D., Ozdaglar, A., and Tahbaz-Salehi, A. (2015). Systemic risk and stability in financial networks. *American Economic Review*, 105(2):564–608.
- Anand, K., Craig, B., and von Peter, G. (2015). Filling in the blanks: network structure and interbank contagion. *Quantitative Finance*, 15(4):625–636.

- Arlot, S. and Celisse, A. (2010). A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79.
- Battiston, S., Delli Gatti, D., Gallegati, M., Greenwald, B., and Stiglitz, J. E. (2012a). Liaisons dangereuses: Increasing connectivity, risk sharing, and systemic risk. *Journal of Economic Dynamics and Control*, 36(8):1121–1141.
- Battiston, S. and Martinez-Jaramillo, S. (2018). Financial networks and stress testing: Challenges and new research avenues for systemic risk analysis and financial stability implications. *Journal of Financial Stability*, 35:6–16.
- Battiston, S., Puliga, M., Kaushik, R., Tasca, P., and Caldarelli, G. (2012b). Debtrank: Too Central to Fail? Financial Networks, the Fed and Systemic Risk. *Scientific Reports*, 2(541).
- Bergmeir, C. and Benítez, J. M. (2012). On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213.
- Blundell, R. and Bond, S. (1998). Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics*, 87:115–143.
- Boccaletti, S., Bianconi, G., Criado, R., del Genio, C., Gómez-Gardeñes, J., Romance, M., Sendiña-Nadal, I., Wang, Z., and Zanin, M. (2014). The structure and dynamics of multilayer networks. *Physics Reports*, 544(1):1–122.
- Boss, M., Elsinger, H., Summer, and Thurner (2004). Network topology of the interbank market. *Quantitative Finance*, 4:677–684.
- Capponi, A. and Chen, P. C. (2015). Systemic risk mitigation in financial networks. *Journal of Economic Dynamics and Control*, 58:152–166.
- Castro Miranda, R. C., Stancato de Souza, S. R., Silva, T. C., and Tabak, B. M. (2014). Connectivity and systemic risk in the Brazilian national payments system. *Journal of Complex Networks*, 2(4):585–613.
- Chen, C., Iyengar, G., and Moallemi, C. C. (2013). An axiomatic approach to systemic risk. *Management Science*, 59(6):1373–1388.
- Csóka, P. and Jean-Jacques Herings, P. (2018). Decentralized clearing in financial networks. *Management Science*, 64(10):4681–4699.
- da F. Costa, L., Rodrigues, F. A., Travieso, G., and Boas, P. R. V. (2005). Characterization of complex networks: A survey of measurements. *Advances in Physics*, 56(1):167–242.
- Decelle, A. (2023). An introduction to machine learning: a perspective from statistical physics. *Physica A: Statistical Mechanics and its Applications*, 631:128154. Lecture Notes of the 15th International Summer School of Fundamental Problems in Statistical Physics.
- Demange, G. (2018). Contagion in financial networks: A threat index. *Management Science*, 64(2):955–970.

- Dunlavy, D. M., Kolda, T. G., and Acar, E. (2011). Temporal link prediction using matrix and tensor factorizations. *ACM Transactions on Knowledge Discovery from Data*, 5(2):10:1–10:27.
- Eisenberg, L. and Noe, T. (2001). Systemic risk in financial systems. *Management Science*, 47(2):236–249.
- Elliott, M., Golub, B., and Jackson, M. O. (2014). Financial networks and contagion. *American Economic Review*, 104(10):3115–53.
- Elsinger, H., Lehar, A., and Summer, M. (2006). Risk assessment for banking systems. *Management Science*, 52(9):1301–1314.
- Erol, S. and Ordoñez, G. (2017). Network reactions to banking regulations. *Journal of Monetary Economics*, 89:51–67.
- Freeman, L. C. (1977). A set of measures of centrality based on betweenness. *Sociometry*, 40(1):35–41.
- Gai, P. and Kapadia, S. (2010). Contagion in financial networks. *Proceedings of the Royal Society*, 4666(2120):2401–2424.
- Gandy, A. and Veraart, L. A. M. (2017). A Bayesian methodology for systemic risk assessment in financial networks. *Management Science*, 63(12):4428–4446.
- Garey, M. R. and Johnson, D. S. (1990). *Computers and Intractability; A Guide to the Theory of NP-Completeness*. W. H. Freeman & Co., New York, NY, USA.
- Haussler, D. (1999). Convolution kernels on discrete structures. Technical Report UCS-CRL-99-10, University of California at Santa Cruz, Santa Cruz, CA, USA.
- Katz, L. (1953). A new status index derived from sociometric analysis. *Psychometrika*, 18(1):39–43.
- Li, J.-C., Xu, M.-Z., Han, X., and Tao, C. (2022). Dynamic risk resonance between crude oil and stock market by econophysics and machine learning. *Physica A: Statistical Mechanics and its Applications*, 607:128212.
- Liben-Nowell, D. and Kleinberg, J. (2007). The link-prediction problem for social networks. *Journal of the American Society for Information Science and Technology*, 58(7):1019–1031.
- Newman, M. E. (2002). Assortative mixing in networks. *Phys. Rev. Lett.*, 89(20):208701.
- Newman, M. E. J. (2003). Mixing patterns in networks. *Phys. Rev. E*, 67(2):026126.
- Nier, E., Yang, J., Yorulmazer, T., and Alentorn, A. (2007). Network models and financial stability. *Journal of Economic Dynamics and Control*, 31(6):2033–2060.
- Rogers, L. C. G. and Veraart, L. A. M. (2013). Failure and rescue in an interbank network. *Management Science*, 59(4):882–898.
- Schölkopf, B., Tsuda, K., and Vert, J.-P. (2004). *Kernel methods in computational biology*. MIT Press, Cambridge, Mass.

- Shervashidze, N., Schweitzer, P., van Leeuwen, E. J., Mehlhorn, K., and Borgwardt, K. M. (2011). Weisfeiler-Lehman graph kernels. *Journal of Machine Learning Research*, 12:2539–2561.
- Silva, T. C., da Silva, M. A., and Tabak, B. M. (2017a). Systemic risk in financial systems: A feedback approach. *Journal of Economic Behavior & Organization*, 144:97 – 120.
- Silva, T. C., Souza, S. R. S., and Tabak, B. M. (2017b). Monitoring vulnerability and impact diffusion in financial networks. *Journal of Economic Dynamics and Control*, 76:109–135.
- Silva, T. C. and Zhao, L. (2012). Network-based high level data classification. *IEEE Transactions on Neural Networks and Learning Systems*, 23(6):954–970.
- Silva, T. C. and Zhao, L. (2015). High-level pattern-based classification via tourist walks in networks. *Information Sciences*, 294(0):109 – 126.
- Silva, T. C. and Zhao, L. (2016). *Machine learning in complex networks*. Springer.
- Souza, S., Silva, T., Tabak, B., and Guerra, S. (2016). Evaluating systemic risk using bank default probabilities in financial networks. *Journal of Economic Dynamics and Control*, 66:54–75.
- Tabak, B. M., Silva, T. C., Fiche, M. E., and Braz, T. (2021). Citation likelihood analysis of the interbank financial networks literature: A machine learning and bibliometric approach. *Physica A: Statistical Mechanics and its Applications*, 562:125363.
- Tasca, P., Battiston, S., and Deghi, A. (2017). Portfolio diversification and systemic risk in interbank networks. *Journal of Economic Dynamics and Control*, 82:96–124.
- Wang, P. and Zong, L. (2023). Does machine learning help private sectors to alarm crises? evidence from china’s currency market. *Physica A: Statistical Mechanics and its Applications*, 611:128470.